Pointing Corrections on Rho-OphA

Date: August 14, 2025

From: Nathan Scott

Currently, pointing is corrected using calibration targets with known sky coordinates like quasars. This is done with observations adjacent to science targets with pointing offsets calculated for each in the Az/El frame. Corrections are made for each science observation using a linear interpolation based on time. This method works well for aligning source observations relative to each other to produce an unblurred product but as I show in this memo, there may still be an absolute offset in RA/DEC as well as potential for improvement in relative alignment which can lead to higher fluxes in the coadded map.

The data I used came from Rho OphA and consisted of 7 files for science numbered 131947, 131949, 134710, 134712, 134714, 134859, and 134861. These approximately 20 minute measurements are accompanied by the calibration observations 131946, 131948, 131950, 134709, 134711, 134713, 134715, 134858, 134860, 134862. First, the paths in the 72_reduce.yaml file corresponding to pointing were removed leading to expectedly poor absolute pointing for each file. Reductions were run using 3 eigenmodes of PCA cleaning with FRUITLoops disabled. No other notable changes to configuration files were made. I selected the Young Stellar Object VLA 1623-243, the most prominent point source in the map, as the reference point with accurate coordinates RA: 246.61008333°, Dec: -24.40833333° (Bibcode: 2009ApJS..181..321E) queried from Simbad. For each observation, the WCS coordinates for VLA 1623 in ICRS were converted to a pixel location and the image was cropped to 40" by 40" centered on the true coordinate. Visual inspection of the location of point sources showed offsets in the range of a few arcseconds and depended greatly on the day of observation. The point source is located among gas and dust so a flux cut was made at 200 MJy/beam which effectively isolated the source, although there were slight deviations from rotational symmetry on a few observations. A 2D Gaussian was fit to the object restricted to a circular profile and the center was visualized showing good Gaussian fits located in the middle of the PSF. I converted it back to WCS coordinates then subtracted the true coordinate to get the Right Ascension and Declination offsets. Citlali takes offsets in the Az/El frame requiring a coordinate transformation using a simple rotation around the median Parallactic Angle. Whether this rotation was clockwise (CW) or counterclockwise (CCW) was determined experimentally. The CW rotation from the equatorial frame to the Az/El frame was done using Equation 1.

$$\begin{bmatrix} \Delta Az \\ \Delta El \end{bmatrix} = \begin{bmatrix} \cos(PA) & \sin(PA) \\ -\sin(PA) & \cos(PA) \end{bmatrix} \begin{bmatrix} \Delta RA \\ \Delta DEC \end{bmatrix} \tag{1}$$

These new coordinate offsets were then inputted to the 72_reduce.yaml configuration file with epoch set to 0. The reduction was then run again with offsets corresponding to a CW and CCW rotations. Finally, the same code was run once again to evaluate new offsets as well as to visualize improvements. Updated pointing offsets show no errors greater than 1 arcsecond.

The figures below illustrate the effects of these offsets. Assuming correct WCS information for VLA 1623-243, it's clear the CW offset has the best absolute and relative positioning as the point source sits at or very close to the true coordinates for each observation. The CCW offset is included as a point of reference showing that the CW rotation is correct. For the coadded map, Figure 1, we see that pointing improvements increase peak flux in the point source since there is less blurring between observations. Just by adjusting the relative pointing, we can increase the peak flux in the coadd from 944 MJy/beam to 1050 MJy/beam which is an 11% increase.

This method seems to improve both absolute and relative pointing offsets compared to the previous method. Using calibration observations, the relative offsets are very good meaning the problem may lie in the point sources chosen as pointing calibrators, although to be certain would require additional testing with other sources. Regardless, this new method shows promise as a way to improve beam size and produce more accurately pointed maps. Best pointing offsets corresponding to the CW rotation are given in Table 1.

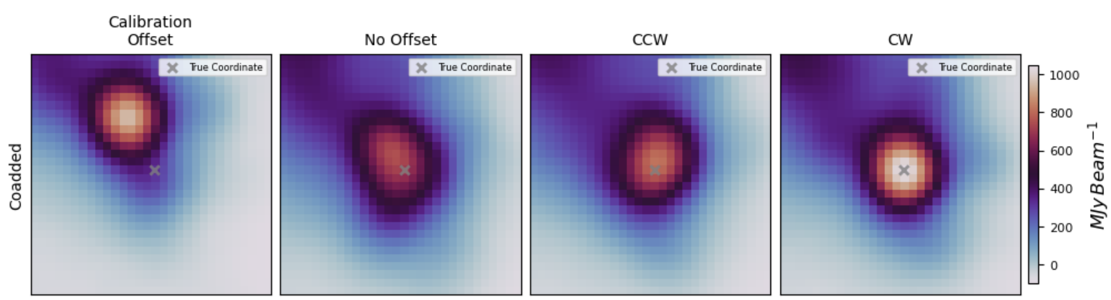

Figure 1: Maps showing differences in positioning relative to VLA 1623-243 for the coadded map with dimensions 30" x 30". Left to right: Default offsets using calibration observations, no pointing offsets, counterclockwise rotation, clockwise rotation.

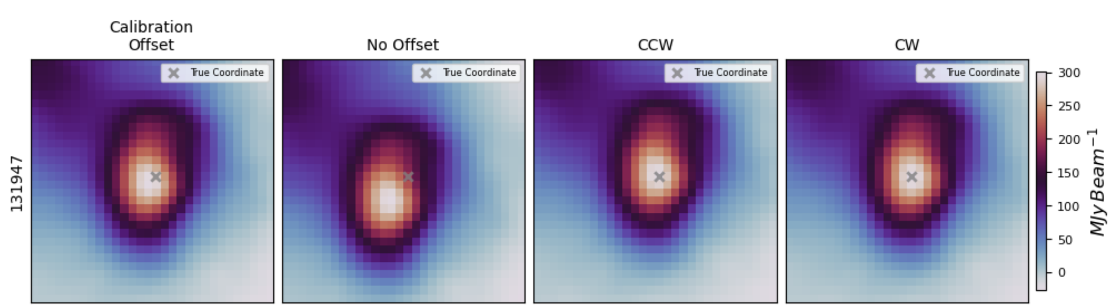

Figure 2: Same as Figure 1 except for 131947

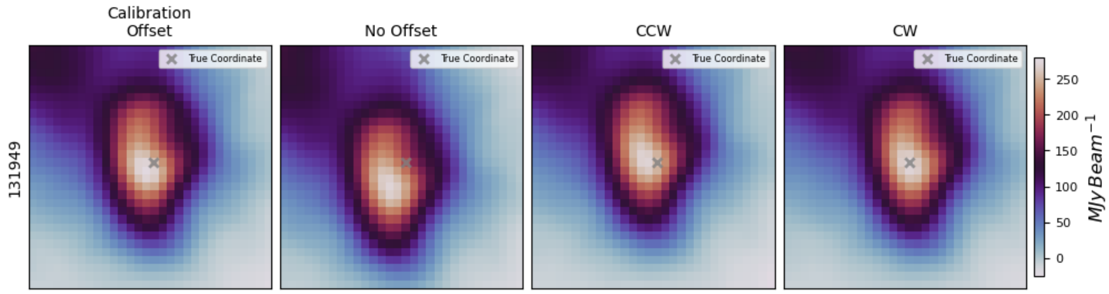

Figure 3: Same as Figure 1 except for 131949

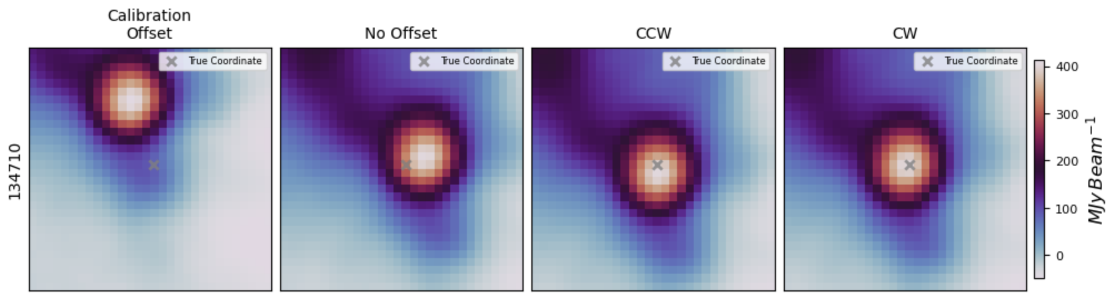

Figure 4: Same as Figure 1 except for 134710

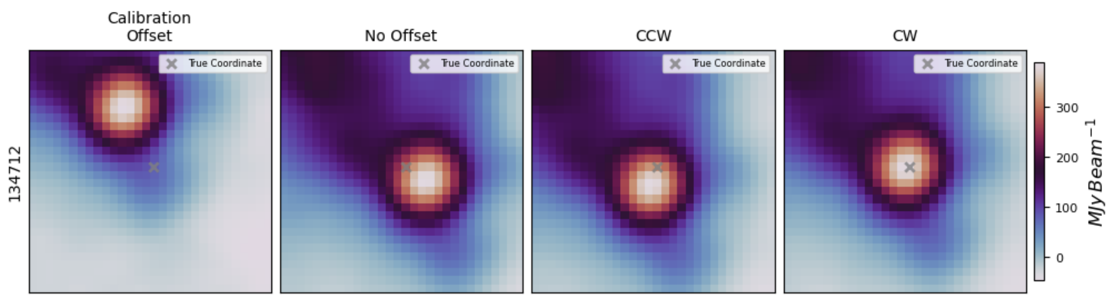

Figure 5: Same as Figure 1 except for 134712

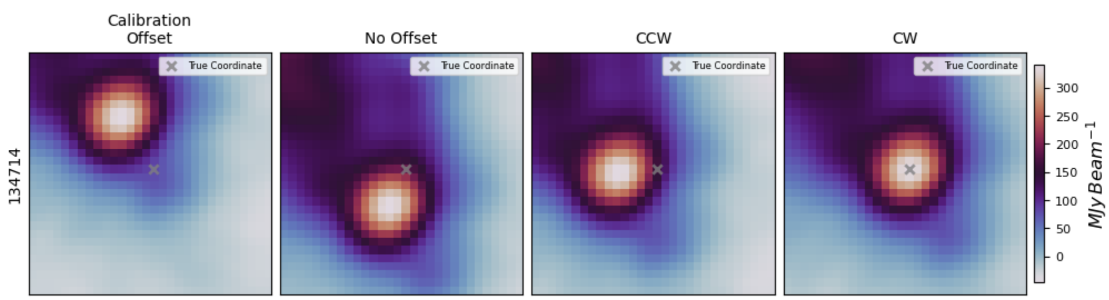

Figure 6: Same as Figure 1 except for 134714

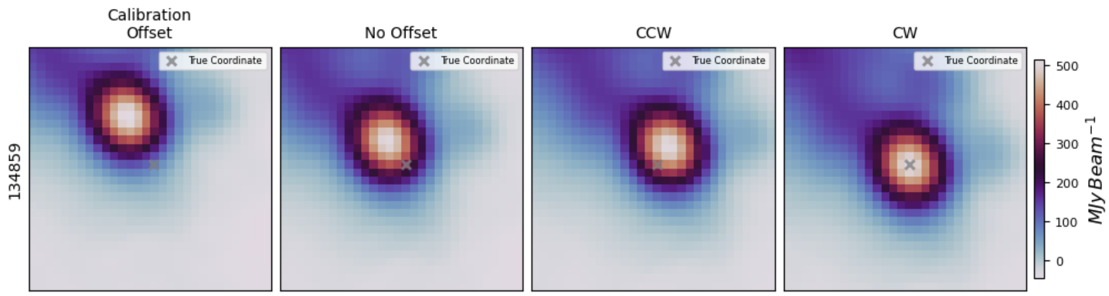

Figure 7: Same as Figure 1 except for 134859

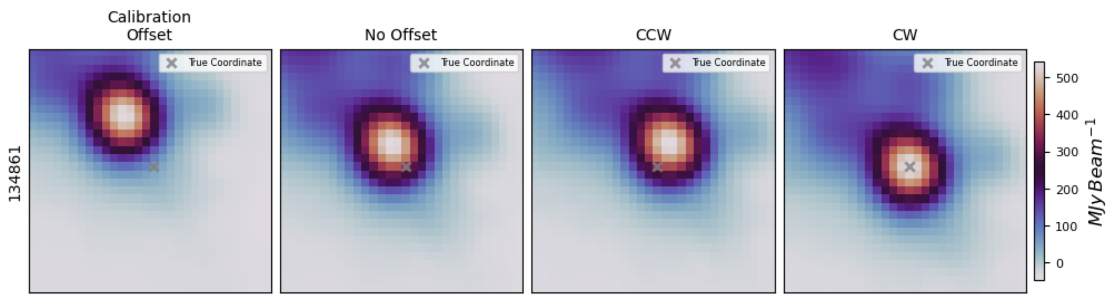

Figure 8: Same as Figure 1 except for 134861

Observation	Az Offset (")	El Offset (")
131947	2.59	2.66
131949	2.58	2.11
134710	-2.83	-0.53
134712	-2.18	2.44
134714	3.97	2.97
134859	1.18	-3.53
134861	0.19	-3.22

Table 1: Updated offsets used in Citlali corresponding to a CW rotation matrix